Housing machining line

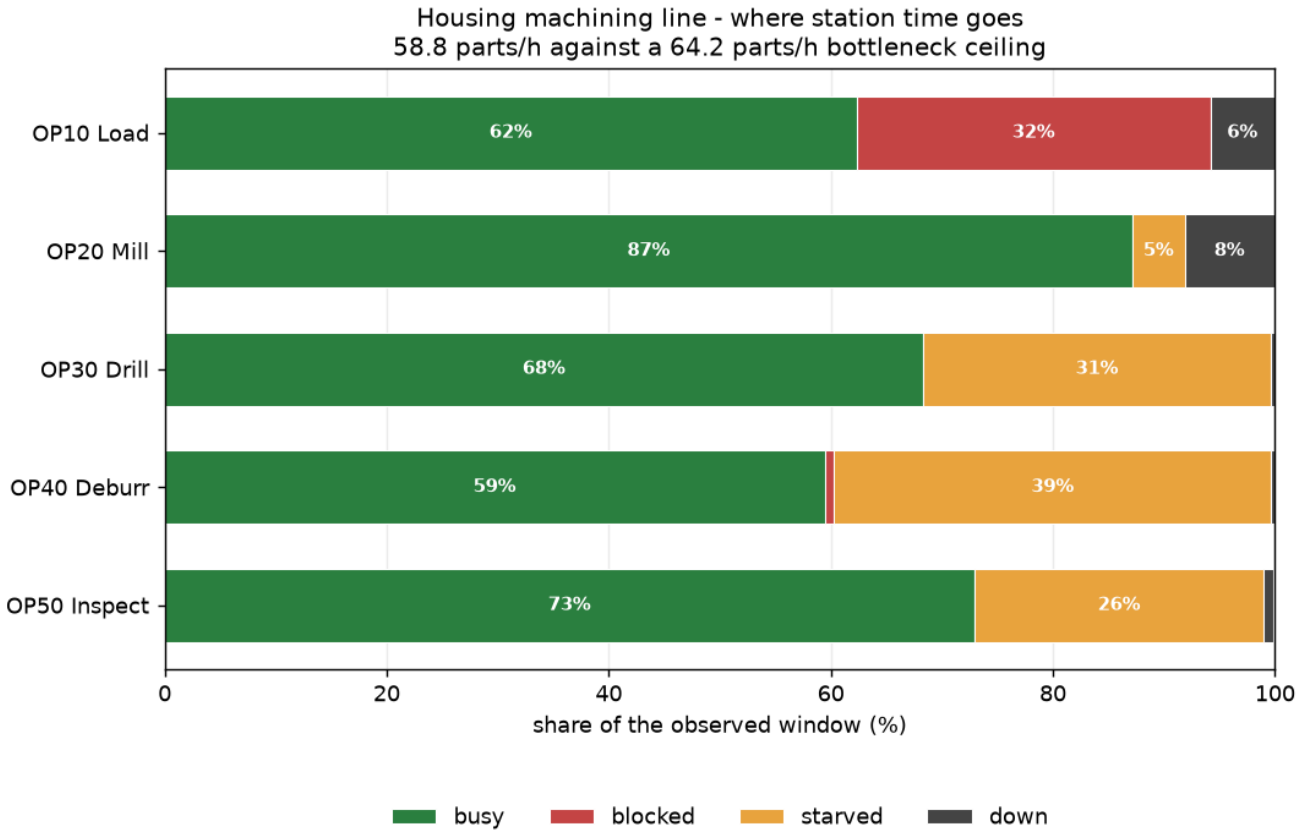
Discrete-event simulation · 5 stations · buffers [2, 2, 2, 2] · 8 h observed after a 1 h warm-up · generated 2026-11-03 11:12

58.8 parts an hour against a 64.2 ceiling

Line efficiency 91.5% of the ceiling set by OP20 Mill. Mean WIP 6.2 parts, flow time 6.2 minutes.

58.8	64.2	91.5%	6.2
throughput, parts/h	ceiling, parts/h	line efficiency	mean WIP, parts

Little's law check: mean WIP 6.17 parts against throughput × flow time = 6.22 (0.74% apart). This is a conservation identity, not a model output — a run that fails it has broken bookkeeping. The small residual is the parts already in the line when the warm-up window closed.



Where station time goes

No station is overloaded, yet the line makes 58.8 parts/h against a 64.2 parts/h ceiling. The gap is not downtime — the downtime is already inside the ceiling. It is the cost of variability plus finite buffers, and it cannot be fixed by fixing machines. OP40 Deburr idles 39% of the window waiting for work the bottleneck has not released yet.

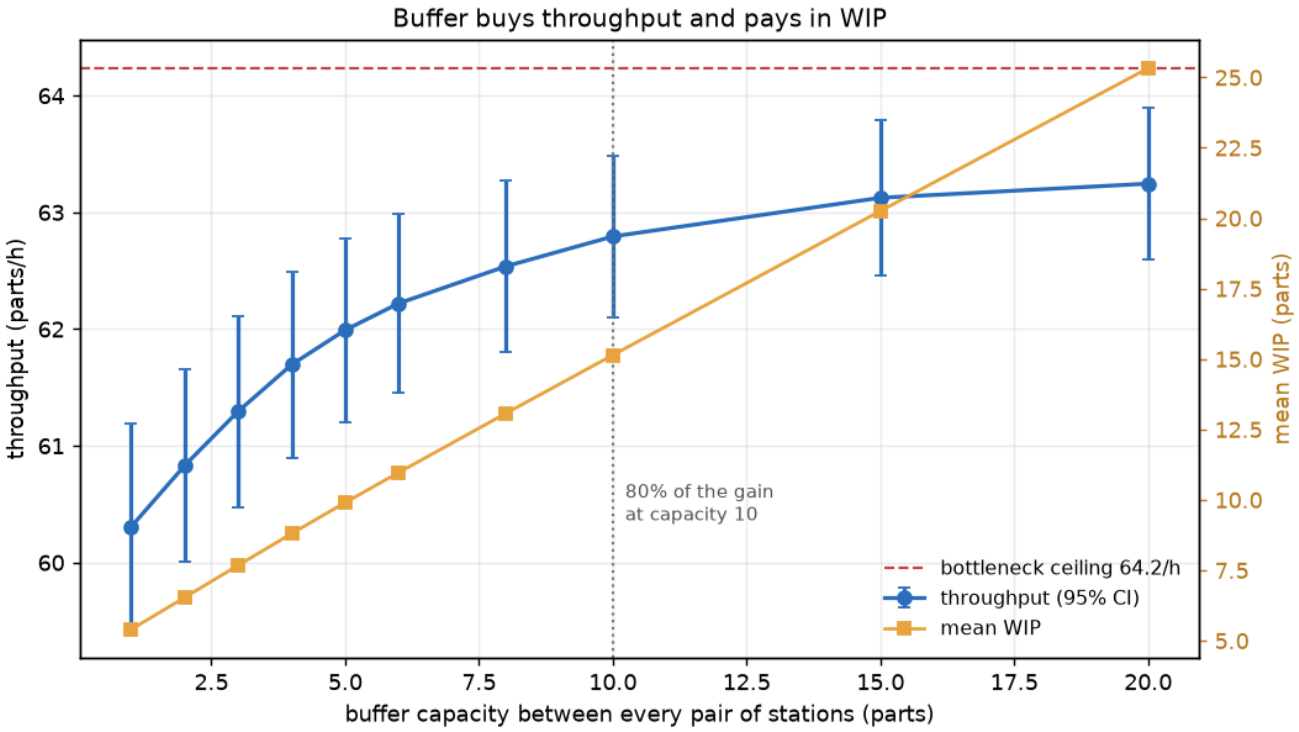
station	busy	blocked	starved	down	failures	parts	waits on
OP10 Load	62.3%	31.8%	0.0%	5.8%	4	479	blocked (downstream full)
OP20 Mill	87.2%	0.0%	4.7%	8.1%	6	480	pace-setter (waits on nobody)
OP30 Drill	68.3%	0.0%	31.3%	0.3%	1	479	starved (upstream empty)
OP40 Deburr	59.4%	0.9%	39.3%	0.3%	2	478	starved (upstream empty)
OP50 Inspect	72.9%	0.0%	26.1%	0.9%	2	478	starved (upstream empty)

Busy, blocked, starved and down are shares of the whole observed window, so they sum to one for each station.

How much buffer?

12 replications per point, 40 h each.

The marginal column is the decision: every extra part of WIP buys less than the one before, and flow time — how long a quality problem stays undetected on the line — rises with it the whole way.

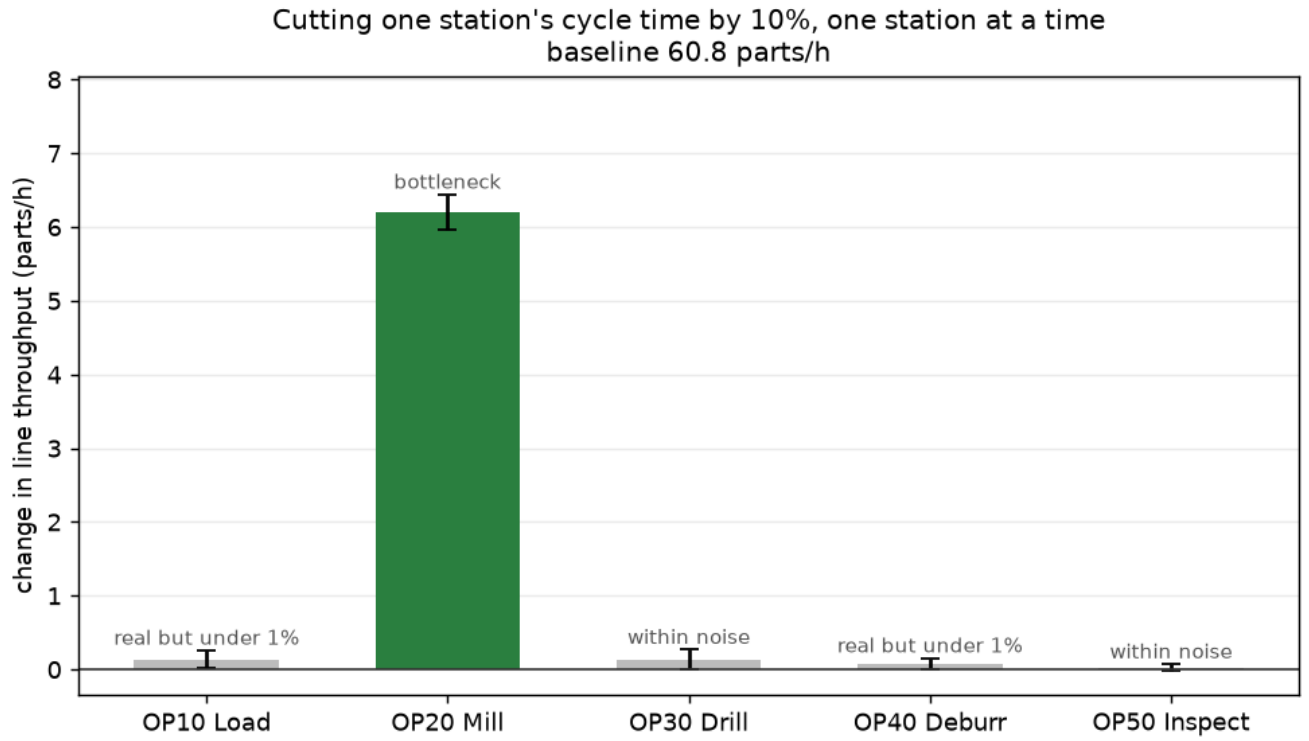


capacity	throughput, parts/h	mean WIP	flow time, min	marginal parts/h per WIP part
1	60.31 ± 0.88	5.40	5.3	—
2	60.83 ± 0.83	6.54	6.3	0.461
3	61.30 ± 0.82	7.68	7.4	0.407
4	61.70 ± 0.80	8.81	8.4	0.354
5	61.99 ± 0.79	9.91	9.4	0.269
6	62.22 ± 0.77	10.97	10.4	0.215
8	62.54 ± 0.74	13.07	12.3	0.151
10 ← knee	62.79 ± 0.69	15.14	14.2	0.124
15	63.12 ± 0.67	20.26	18.9	0.065
20	63.25 ± 0.65	25.33	23.5	0.024

Where does improvement pay?

Baseline 60.83 ± 0.83 parts/h.

The change column is the paired difference: every variant is run on the same seeds, so replication k of the improved line shares every failure time at every station the change did not touch. The shared variation cancels inside each pair, which is why the interval on a change is far tighter than the interval on either line's absolute throughput. "Within noise" means that paired interval still contains zero.

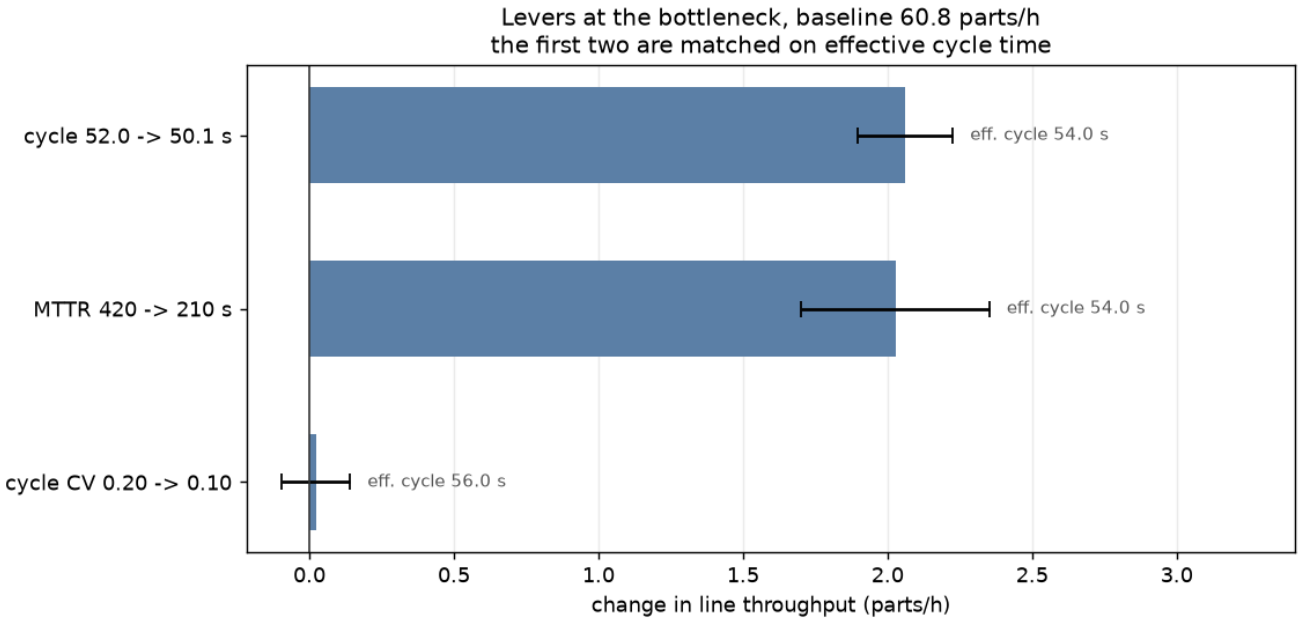


station made 10% faster	throughput, parts/h	paired change	verdict
OP10 Load	60.97 ± 0.87	+0.14 ± 0.12	real but under 1%
OP20 Mill ← bottleneck	67.03 ± 0.98	+6.20 ± 0.24	material
OP30 Drill	60.96 ± 0.79	+0.13 ± 0.14	within noise
OP40 Deburr	60.91 ± 0.83	+0.08 ± 0.07	real but under 1%
OP50 Inspect	60.86 ± 0.81	+0.03 ± 0.05	within noise

Speed, reliability or variability?

Baseline 60.83 ± 0.83 parts/h.

The first two levers are matched on effective cycle time, so a capacity calculation would price them identically. The third does not change effective cycle time at all — cycle variability is invisible to that calculation, which is precisely why it gets overlooked.



lever at OP20 Mill	effective cycle, s	throughput, parts/h	paired change	verdict
cycle 52.0 -> 50.1 s	54.02	62.89 ± 0.90	+2.06 ± 0.17	material
MTTR 420 -> 210 s	54.02	62.86 ± 0.62	+2.03 ± 0.33	material
cycle CV 0.20 -> 0.10	56.04	60.85 ± 0.82	+0.02 ± 0.12	within noise

Seeded throughout, so re-running reproduces every figure and every interval above.